

2.3.3 Inverse Functions and Compositions of Functions

Definition 9: Let f be a one-to-one correspondence from the set A to the set B . The inverse function of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a)=b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b)=a$ when $f(a)=b$.

Remark: Be sure not to confuse the function f^{-1} with the function $1/f$, which is the function that assigns to each x in the domain the ~~value~~ value $1/f(x)$. Notice that the latter makes sense only when $f(x)$ is a nonzero real number.

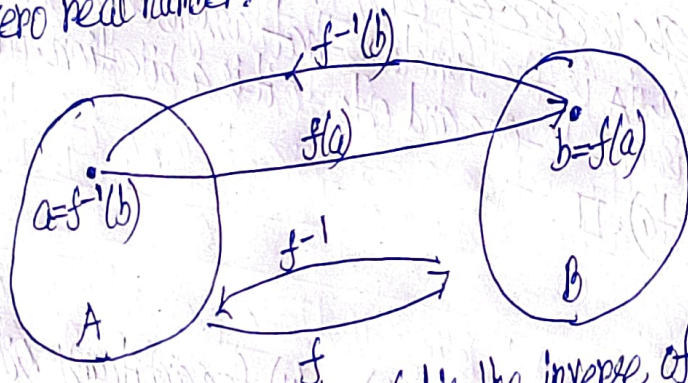


FIGURE 6: The function f^{-1} is the inverse of function f .

A one-to-one correspondence is called invertible because we can define an inverse of this function. A function is not invertible if it is not a one-to-one correspondence, because the inverse of such a function does not exist.

Example 19: Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a)=2$, $f(b)=3$ and $f(c)=1$. Is f invertible, and if it is, what is its inverse?

Solution: It's easy to see that $\forall x, y$, where $x, y \in \{a, b, c\}$ if $f(x)=f(y)$, then $x=y$. By Definition 5, f is one-to-one. $\forall z, \exists y$ where $z \in \{1, 2, 3\}$ and $y \in \{a, b, c\}$ s.t. $f(y)=z$. This too is easy to verify. By Defn. 7, f is onto. $\therefore$ From Defn. 8, f is bijective and has a one-to-one correspondence. By Defn. 9, f has an inverse. f^{-1} is defined from $\{1, 2, 3\}$ to $\{a, b, c\}$ s.t.